

TalentMind AI

Enterprise HR Knowledge & Talent Intelligence Platform

Project Charter & Problem Definition Document

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1. Project Overview

TalentMind AI is an AI-powered Enterprise Knowledge and Talent Intelligence Platform designed to demonstrate modern Retrieval-Augmented Generation (RAG) architectures, semantic search, hybrid retrieval, metadata filtering, retrieval agents, document versioning, and production-grade evaluation techniques.

The purpose of this project is to provide a hands-on implementation experience covering concepts frequently assessed in AI Engineer interviews while simulating a realistic enterprise use case.

This project will act as:

- AI Engineer Portfolio Project
 - RAG Learning Lab
 - Interview Preparation Project
 - Production Architecture Reference Implementation
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2. Business Problem

Organizations maintain thousands of documents, policies, resumes, and knowledge assets across multiple systems.

Employees and recruiters face challenges such as:

Knowledge Silos

Information exists in:

- PDFs
- SharePoint
- Google Drive
- Internal Wikis
- HR Portals

Finding relevant information is time-consuming.

Resume Search Challenges

Recruiters manually review resumes.

Keyword-based systems fail to identify semantically similar profiles.

Example:

Query:

Python Developer

Resume:

Backend Engineer
FastAPI
Azure

Traditional keyword search may fail to identify this candidate.

Repetitive HR Queries

Common employee questions:

- Can I work remotely?
- What is my medical coverage?
- What is the travel allowance?
- How many annual leaves are provided?

HR teams spend significant time answering repetitive queries.

Hallucination Risks

Standalone LLMs may generate incorrect answers if they lack access to enterprise knowledge.

A grounded retrieval mechanism is required.

3. Project Objectives

The project aims to demonstrate the implementation of:

- ✓ Embeddings
- ✓ Semantic Search
- ✓ Similarity Search
- ✓ Vector Databases
- ✓ Vector Stores
- ✓ Metadata Filtering
- ✓ Chunking Strategies
- ✓ Retrieval Pipelines
- ✓ Basic RAG
- ✓ Hybrid Retrieval
- ✓ Retrieval Agents
- ✓ RAG Evaluation
- ✓ Latency Optimization
- ✓ Document Versioning
- ✓ Hallucination Reduction

4. Dataset Requirements

The project should utilize realistic datasets.

Dataset 1 – HR Policy Documents

Purpose:

Enterprise Knowledge Base

Examples:

- Leave Policies
- Travel Policies
- Medical Benefits
- Remote Work Guidelines

Format:

PDF

DOCX

TXT

Recommended Sources:

Public HR manuals

Synthetic HR policies generated using GPT

Dataset 2 – Resume Dataset

Purpose:

Talent Search

Matching

Recommendations

Recommended Fields:

Candidate Name

Title

Experience

Skills

City

Nationality

Certifications

Education

Recommended Size:

500–5,000 resumes

Recommended Source:

Kaggle Resume Datasets

Synthetic Resume Generation

Dataset 3 – Job Descriptions

Purpose:

JD Matching

Examples:

Backend Engineer

Data Scientist

AI Engineer

HR Executive

Recommended Size:

200–500 JDs

Dataset 4 – Employee Records

Purpose:

Agentic Retrieval

Fields:

Employee ID

Leave Balance

Appraisal Summary

Department

Manager

Dataset Type:

CSV

PostgreSQL

5. Expected Outputs

The system should be capable of producing:

Output 1

Policy Answers

Example:

Question:

Can I work remotely?

Answer:

Employees may work remotely up to three days per week subject to manager approval.

Source Citation:

HRPolicy.pdf

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Output 2

Candidate Recommendations

Example:

Ahmed Khan

Match Score:

89%

Reasons:

FastAPI

Azure

Dubai

6 years experience

Output 3

Agentic Responses

Question:

Show Ahmed's leave balance and appraisal comments.

System Response:

Leave Balance: 12 Days

Appraisal Summary:

Exceeded Expectations

Source:

Employee Database

6. Deliverables

At project completion, the following deliverables are expected.

Deliverable 1

FastAPI Backend

Deliverable 2

Qdrant Collection

Deliverable 3

PostgreSQL Database

Deliverable 4

Document Ingestion Pipeline

Deliverable 5

Policy Assistant

Deliverable 6

Talent Search Engine

Deliverable 7

Retrieval Agent

Deliverable 8

Evaluation Dashboard

Deliverable 9

GitHub Repository

Deliverable 10

Architecture Document

Deliverable 11

Demo Video

7. Interview Questions Covered

By implementing this project, the engineer should be able to confidently answer questions related to:

- How embeddings work
 - Metadata filtering
 - Hybrid Retrieval
 - Chunking strategies
 - Latency optimization
 - Hallucination reduction
 - File updates and versioning
 - Retrieval Agents
 - RAG evaluation
 - Production scaling
 - Vector DB selection
 - Embedding model selection
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8. Success Criteria

The project will be considered successful if:

- Policy retrieval latency is under 2 seconds.
- Retrieval precision exceeds 90%.
- Faithfulness exceeds 95%.
- Updated files are automatically re-indexed.
- Retrieval agents can combine information from multiple sources.
- The project demonstrates all major RAG concepts discussed in AI Engineer interviews.